Example 3.12 If one worker earns ₹200 per day, how many workers can be employed within a monthly budget of ₹3 Lakh?

Solution

Let the number of workers be x.

Then, the wages that x workers will earn per day = ₹$200x$

The wages that x workers will earn per month= ₹$(200x \times 30)$= ₹$6000x$

Given that, this amount cannot exceed ₹300000.

Otherwise, it can be written as $6000x \leq 300000$

$$\frac{6000x}{6000} \leq \frac{300000}{6000} \quad \text{[Divide by 6000 on both sides]}$$

$$x \leq 50$$

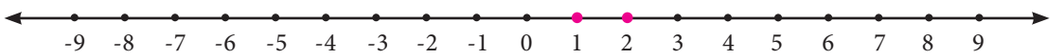
Thus, up to 50 workers can be employed on a monthly budget of ₹300000.

3.5.2 Graphical representation of Inequation

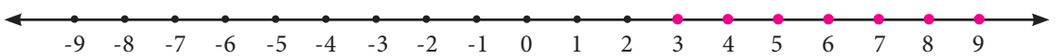
The solutions of an inequation can be represented on the number line by marking the true values of solutions with different colour on the number line.

Look into the following inequations and its graphical representation on number line. Here, we consider the solution belongs to natural numbers. That is, each and every value of the solution is a natural number.

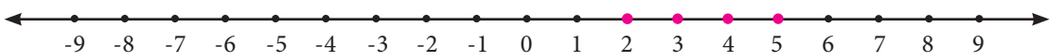
1. When $x < 3$, the solution in natural numbers are 1 and 2. Its graph on number line is shown below:



2. When $x \geq 3$, the solutions are natural numbers 3, 4, 5, ... and its graph is as shown below:



3. To mark the values represented by the inequation $2 \leq x \leq 5$, the solutions are set of natural numbers 2, 3, 4 and 5 and its graph is as given below:



Example 3.13 Represent the solutions $-8 < 2x < 10$ in a number line, where x is a natural number.

Solution

$-8 < 2x < 10$

$$\frac{-8}{2} < \frac{2x}{2} < \frac{10}{2} \quad \text{[Dividing the inequation by 2]}$$

$-4 < x < 5$

Note

Since the solution is restricted to natural numbers, -3, -2, -1 and 0 have not been marked as solutions.

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